

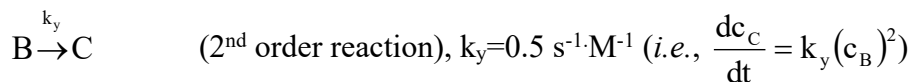
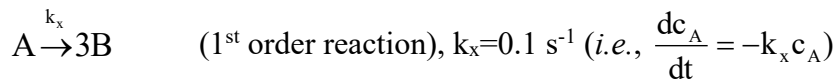
PROBLEM SET 9

(Due Apr 7, 2023 @ 8:30 AM in Gradescope and Moodle)

This problem set is to be completed **individually** all in **Excel**. You will submit your own Excel assignments via Moodle and attach screenshots to your Gradescope submission. Make sure you start a blank workbook on your own computer. Do not forget to press the submit button in Moodle.

Problem 1 (40 points total)

The following chemical reactions take place in a liquid-phase batch reactor of constant volume V .



The reactor is initially charged with a solution of A having a concentration of 0.1 mol/L. No B and C species are present at time=0 seconds.

In HW8, we have already figured out the differential equations for dc_A/dt , dc_B/dt , and dc_C/dt (see quoted below).

“Using the order of chemical reaction we learned in the class, derive differential equations of dc_A , dc_B , and dc_C as a function of dt (e.g., $dc_A/dt=$, $dc_B/dt=$, and $dc_C/dt=$). Plug in known variables (e.g., k_x and k_y) in your differential equations. Provide the initial conditions for each differential equation.”

Now, let's continue: use the expressions you previously derived to answer **Part a-c**.

- (25 points)** Solve the differential equations derived in HW8 using the 4th order Runge-Kutta method on the time interval $t \in [0, 80]$ seconds. Provide the answers in the Excel sheet; mark clearly the cells that contain the time, c_A , c_B , and c_C values. Use the “time step” value in the Runge-Kutta method (i.e., $h = t_{i+1} - t_i$) not larger than 0.5 second.
- (10 points)** Plot c_A , c_B , and c_C and the total concentration ($=c_A+c_B+c_C$) as a function of time on a single plot.
- (5 points)** From the plot in part b) you will see that c_B initially increases, reaches a maximum at some intermediate time, and then decreases. At what time (starting from the beginning of the process) does c_B reach the maximum value? What is the value of c_B at this point? Find your answer to this part numerically based on your results for part b).

Problem 2 (30 points total)

Consider the following ordinary differential equation (ODE) on the interval of $m \in [1, 3]$:

$$\frac{dz}{dm} - (2z - m) = \ln(z^m) \quad \text{initial condition: } z=1 \text{ at } m=1 \quad (1)$$

- (10 points)** Find the solution to the ODE given by Eq. (1) numerically by using the 2nd order Runge-Kutta method by using step increments in m (i.e., $h=m_{i+1}-m_i$) equal to 0.1, 0.05, and 0.02. Plot the results of z as a function of m in the same plot using blue ($h=0.1$), red ($h=0.05$) and green ($h=0.02$) lines.
- (10 points)** Find the solution to the ODE given by Eq. (1) numerically by using the 4th order Runge-Kutta method by using step increments in m (i.e., $h=m_{i+1}-m_i$) equal to 0.1, 0.05, and 0.02. Plot the results of z as a function of m in the same plot using blue ($h=0.1$), red ($h=0.05$) and green ($h=0.02$) lines.
- (10 points)** How easy/difficult is to solve the ODE given by Eq. (1) analytically? Can you solve it? If yes, please provide your solution and submit as a hardcopy. If not, outline your method of solving and comment why solving the ODE is difficult.

Problem 3 (30 points total)

Use the data given in Table 1 (see also attached Excel Workbook that lists the value from Table I – so that you do not have to type those in) above to perform the following tasks:

- (10 points)** In a spreadsheet called “Problem 3a-b”, plot the data in Excel as $y=y(x)$ and use Trendline polynomial of the 3rd order to find the analytical form of the $y=y(x)$ dependence. From the Trendline equation determine the analytical solution for y' (i.e., $y'=dy/dx=f(x)$).
- (10 points)** In a spreadsheet called “Problem 3a-b”, calculate numerically and plot the 1st derivatives of y with respect to x (i.e., y') using the “forward”, “backward” and “central” methods. Comment on the accuracy of each method by comparing your numerical results to the analytical solution.
- (10 points)** In a spreadsheet called “Problem 3c”, find the approximate integral of the data in Table 1 on the interval $x \in [1, 2]$ using the i) trapezoidal rule; ii) Romberg integration; and iii) Simpson 1/3 rule (look at the lecture notes for details about the integration methods). Compare the numerical results to that obtained from the analytical solution. To obtain the analytical solution, integrate the equation you obtain from the Trendline in part a).

Table 1. Data for Problem 3

x	y		x	y
0	2		1.05	1.015625
0.05	2.128125		1.1	1.065
0.1	2.215		1.15	1.151875
0.15	2.264375		1.2	1.28
0.2	2.28		1.25	1.453125
0.25	2.265625		1.3	1.675

0.3	2.225		1.35	1.949375
0.35	2.161875		1.4	2.28
0.4	2.08		1.45	2.670625
0.45	1.983125		1.5	3.125
0.5	1.875		1.55	3.646875
0.55	1.759375		1.6	4.24
0.6	1.64		1.65	4.908125
0.65	1.520625		1.7	5.655
0.7	1.405		1.75	6.484375
0.75	1.296875		1.8	7.4
0.8	1.2		1.85	8.405625
0.85	1.118125		1.9	9.505
0.9	1.055		1.95	10.70188
0.95	1.014375		2	12
1	1			